

Random Variables, Describing Data

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News

- Changing lecture/workshop times non-trivial and unlikely to happen.
- There is a zoom link for every lecture and workshop. Not full hybrid.
- Workshop problems are on canvas week 2 page.
- Assignment 1 is on canvas Assignment & Guidance page
- I made a slack channel for those who want it.
- I made an anonymous google form for PIPS times.
- Python/marimo issues: please ask your fellow students.
- Attendance etc, please email: fosem-info@sussex.ac.uk

Following with Python

Install marimo as per instructions on canvas: Resources & Prerequisites

<MINTED>

Download the notebook from the canvas page for this week's material:

[RandomVariables.py](#)

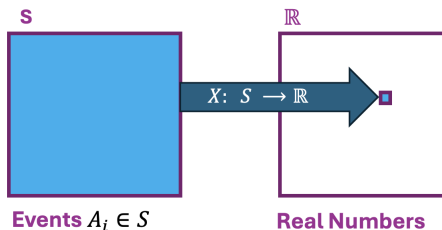
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Data

- Random Variables
- Mean & Expected Value
- Variance & Standard Deviation
- Weighted Mean & Variance
- Other Definitions

What is a Random Variable?

A **Random Variable (RV)** is a **function** that maps events to real numbers. $X(A_i) = x_i$ for all $A_i \in S$



X is the RV: a function

A_i is an event in the sample space S of all possible events.

x_i is the value corresponding to this event, according to X .

Random Variables: Example of Height

The **RV** X is the function that turns the abstract concept of height into a number and unit.

$$X(\text{abstract concept of Lily's height}) = 1.71 \text{ m}$$

$$X(A_i) = x_i \text{ for all } A_i \in S$$

A_i is a height from the sample space S of all possible heights.

x_i is the value corresponding to this height, according to X .

General Notation used by me in this module

Z : the number of Random Variables under consideration

$X = \{X_{(1)}, X_{(2)}, \dots, X_{(Z)}\}$ the Z Random Variables.

$x = \{x_{(1)}, x_{(2)}, \dots, x_{(Z)}\}$ the Z Datasets

$x_{(j)} = \{x_{j,1}, x_{j,2}, \dots, x_{j,N}\}$ is a Dataset of N Measurements

If we have one RV, I will say that the Dataset is one-dimensional (1D) and X is the single RV. Almost all examples in this topic are 1D.

General Notation: RVs

Z : the number of Random Variables under consideration

$X = \{X_{(1)}, X_{(2)}, \dots, X_{(Z)}\}$ **the Z Random Variables.**

$x = \{x_{(1)}, x_{(2)}, \dots, x_{(Z)}\}$ the Z Datasets

$x_{(j)} = \{x_{j,1}, x_{j,2}, \dots, x_{j,N}\}$ is a Dataset of N Measurements

Note that the RVs are denoted by the symbol X no matter whether there is one or fifty of them. I do not use the vector notation eg $\vec{X}$. Assume all X are sets of Z $X_{(j)}$ s.

General Notation: Datasets

Z : the number of Random Variables under consideration

$X = \{X_{(1)}, X_{(2)}, \dots, X_{(Z)}\}$ the Z Random Variables.

$x = \{x_{(1)}, x_{(2)}, \dots, x_{(Z)}\}$ **the Z Datasets**

$x_{(j)} = \{x_{j,1}, x_{j,2}, \dots, x_{j,N}\}$ is a Dataset of N Measurements

Each dataset is denoted $x_{(j)}$ where the subscript (j) identifies which RV the measurements are for. The brackets around the datasets' RV identifiers are there to remind us that $x_{(j)}$ is not a number but a set of them. This is needed because when we see x_j our brains say 'number'.

General Notation: Measurements

Z : the number of Random Variables under consideration

$X = \{X_{(1)}, X_{(2)}, \dots, X_{(Z)}\}$ the Z Random Variables.

$x = \{x_{(1)}, x_{(2)}, \dots, x_{(Z)}\}$ the Z Datasets

$x_{(j)} = \{x_{j,1}, x_{j,2}, \dots, x_{j,N}\}$ **is a Dataset of N Measurements**

I like to use the identifiers $i, \dots, N$ for measurements, and $j, \dots, Z$ for RVs (or sometimes for experiments - more later).

Notation Example

Say we are measuring the heights and weights of a sample of 10 people:

$$Z = 2$$

$X = \{H, W\}$ are the RVs: H is height and W is weight.

$x = \{h, w\}$ are the datasets: h are the measured height values and w are the measured weight values.

$h = \{h_1, h_2, \dots, h_{10}\}$ is a dataset of the $N = 10$ height measurements h_i .

Notice that x is a matrix with 2 rows and 10 columns

Continuous RVs

A **Continuous Random Variable (CRV)** is one for which the measurements can take any value (such as height and weight).

Note that in the real world, measurements always have a finite precision. So the measurements x of a CRV X will not strictly be continuous.

What is true in the real world is also in true in your computer. The precision with which python generates CRVs on my machine is 'float64': precision 15.

A continuous RV is still continuous even though its measurements cannot be.

Discrete RVs

A **Discrete Random Variable (DRV)** is one for which the measurements can only take certain values (such as a roll of dice).

Discrete RVs don't have to be integers, but they do have to be countable.

For example, the probability of dice rolls (any number of dice) are not integers but are Discrete. As I increase the number of dice and/or rolls, I can generate lots of probability values from the original set, but there will always be real numbers I cannot generate (gaps).

Distribution Functions

A **Distribution Function** $f(X; \theta)$ is a theoretical object that describes the True Distribution of the RVs X .

Recall, the RV is a function that returns a real value. So the distribution of X is the distribution of all the possible values that could be returned by X .

The true parameters are denoted by θ .

For example, a straight line:

$f(X; m, c) = mX + c$ is written

$$f(X; \theta) = \theta_1 X + \theta_0$$

If there is only one parameter, and $f(X; \theta)$ is a familiar function such as exp or norm (later!) then we often use a symbol for the parameter to reflect that such as μ or λ .

Distribution Function (DF) as Probability (PDF)

Any DF of a continuous RV can be turned into a Probability DF. For example:

$f(X; \theta) = x^3$ is a function

1 Write $f(X; \theta) = Ax^3$

2 Specify the 'range', for example $X : [0, 1]$

3 Set integral to one: $\int_0^1 Ax^3 dx = 1$

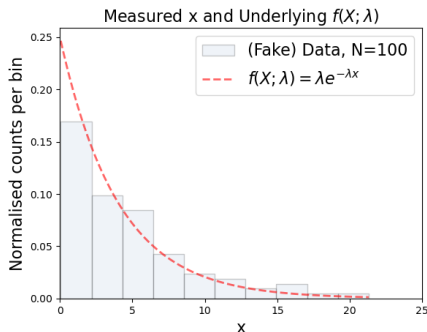
4 Solve for A: $A \frac{1}{4} = 1 \therefore A = 4$

$f_X = 4x^3$ for $0 \leq x \leq 1$ is a valid PDF. Total Probability (Integral) is 1.

The True Distribution

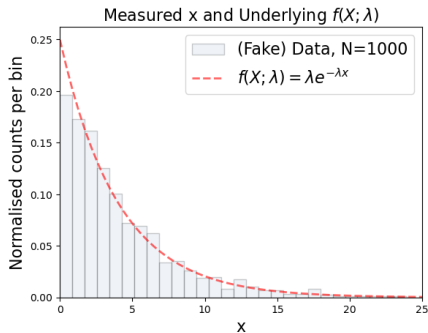
The 1D dataset $x = x_{(1)}$ in this histogram ($N = 100$ measurements) has a familiar shape. Looks like exponential decay! (I cheated - it is fake data)

Drawing the exponential function over the top confirms it is similar.



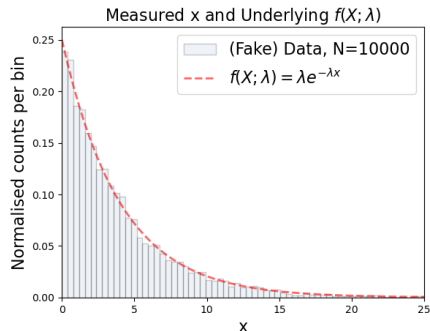
Distribution Functions

If I collect (okay, fake) more data ($N = 1000$), the agreement looks more convincing.



Distribution Functions

Even more data ($N = 10,000$), now they are indistinguishable by eye.



Independent & Identically Distributed RVs

A collection of RVS is **Independent and Identically Distributed (IID)** if:

- 1 They are all mutually independent **and**
- 2 They all have the same underlying probability distribution.

Mutually independent RVs: measuring any of them has no effect on the probabilities of the others.

IID RVs Example

The result of tossing a coin once is a RV $X_{(1)}$. It can have values $x_{(1)} = \{\text{heads}, \text{tails}\}$

The result of tossing the coin again is also a RV, $X_{(2)}$. The same values are possible.

The result of the first toss does not have any effect on the result of the second toss. So the RVs $X_{(1)}$ and $X_{(2)}$ are **Mutually Independent**.

The probability of each toss coming up heads is identical, so the RVs are **Identically Distributed**.

This is true even if the coin is not fair, because it is the same coin being tossed both times.

The Mean of a Dataset

The **Mean** of a 1D dataset ($x = x_{(1)}$) is:

$$\bar{x} = \frac{1}{N} \sum_i^N x_i,$$

where the sum is over the N measurements

The Binned Mean

The **Binned Mean** is important in real life ([why?](#)). It is written:

$$\bar{x} = \frac{1}{N} \sum_i^M n_i c[x]_i$$

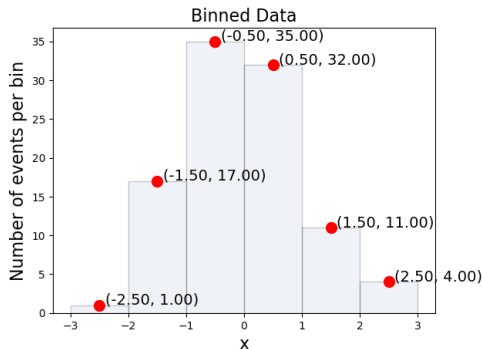
Here n_i is the number of entries in bin i , $c[x]_i$ is the central value of that bin, and the sum is over the M bins rather than the N individual values.

Note that we must still normalise using the number of measurements, N .

The Binned Mean

$$\bar{x} = \frac{1}{N} \sum_i^M n_i c[x]_i$$

Each of the red dots is $(c[x]_i, n_i)$: (bin center, bin count)



The Expected Value

The **Expected Value*** of a single RV $X = X_{(1)}$ is the **True Mean**:

$$E[X] = \sum_i^{\infty} x_i p_i \text{ (X: discrete)}$$

where p_i is the probability of observing the measurement x_i .

If the probabilities of the measurements are described by a *Probability Density Function* f_X (with integral=1), then we can write this:

$$E[X] = \int_{-\infty}^{\infty} x f_X dx \text{ (X:continuous)}$$

Equivalent notation $E[X] \equiv \langle X \rangle \equiv \mu$.

* Also known as the Expectation or Expectation Value.

Aside: The Expected Value for data

The Expected Value of a Dataset of N measurements: $E[x] = \sum_i^N x_i p_i$.

We can use this: what would you **expect** the average number of sixes to be from 100 dice rolls?

Probability is flat $p_i = \frac{1}{6}$ and $x_i = 1$ for a positive outcome (rolled a six)

$$E[x] = \sum_1^{100} 1 \cdot \frac{1}{6} = \frac{100}{6}$$

Note that $E[x]$ is a prediction we can only use when we are certain of the probabilities before hand.

The Expected Value and the Mean

The **Expected Value** of the RV: $E[X] = \sum_i^{\infty} x_i p_i$,

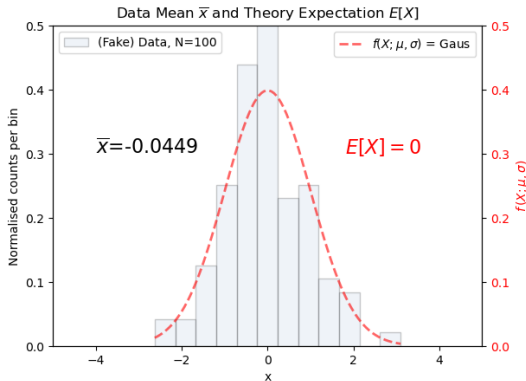
The **Mean** of the Measurements: $\bar{x} = \frac{1}{N} \sum_i^N x_i$,

Observations:

- 1 $E[X]$ sum is to infinity: because in truth, there are ∞ values for X .
- 2 $E[X]$ doesn't have the $\frac{1}{N}$ normalisation: the normalisation is taken care of by p_i .
- 3 $E[X]$ has p_i : p_i is the probability of observing x_i .

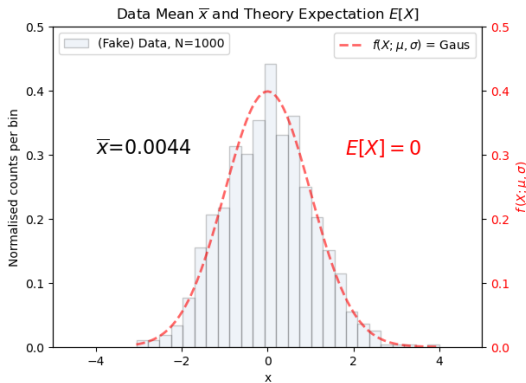
The Expected Value and the Mean

Here is a plot of $N = 100$ measurements (fake data again!) which appear to follow a Gaussian Distribution (I faked them, so they definitely do). The red dashed line is the true distribution f_X of the RV X .



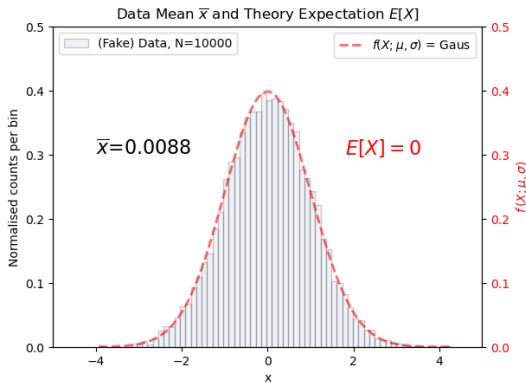
The Expected Value and the Mean

Take more data: $N = 1000$ measurements. The mean has changed, but $E[X]$ has not. It is the True Mean.



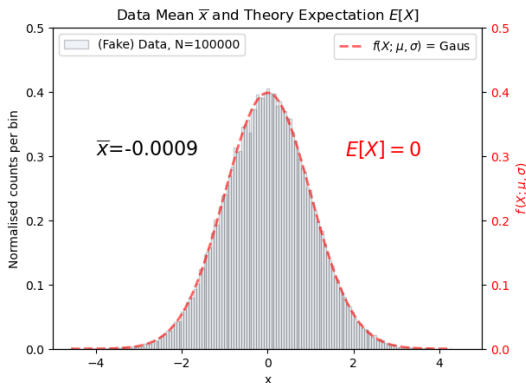
The Expected Value and the Mean

Now $N = 10,000$ measurements. Notice $\bar{x}$ has moved away from $E[X]$ - that can happen.



The Expected Value and the Mean

Now $N = 100,000$ measurements. **In the limit $N \rightarrow \infty$, $\bar{x} \rightarrow E[X]$**



Expected Value Algebra

For RVs A, B :

Expectation values add: $E[A + B] = E[A] + E[B]$

They do not generally multiply: $E[AB] \neq E[A]E[B]$

But, **if A and B are independent, they do multiply:**

$E[AB] = E[A]E[B]$ for A, B independent.

Recall: $E[X] = \int_{-\infty}^{\infty} x f_X dx$

If we can write $f_{XY} = f_X f_Y$, then X, Y are independent.

$E[E[A]] = E[A]$: the expectation is a value, and the expectation of a value is just the value.

Variance

The **Variance** of a set of 1D measurements x is:

$$V[x] = \frac{1}{N} \sum_i^N (x_i - \bar{x})^2$$

This can equivalently be written:

$$V[x] = \overline{x^2} - \bar{x}^2$$

See if you can prove these are equivalent - I will show this some time soon.

The **Standard Deviation** is just the square root of the variance:

$$\sigma_x^2 = V[x]$$

True Variance

The **True Variance** of a RV X is:

$$V[X] = E[X - E[X]]^2$$

Equivalent in mixed notation $V[X] = E[X - \mu]^2$

This can equivalently be written:

$$V[X] = E[X^2] - E^2[X]$$

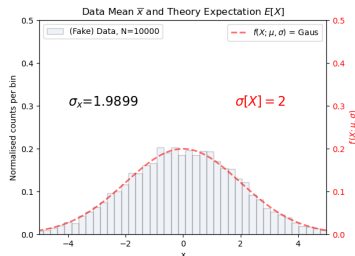
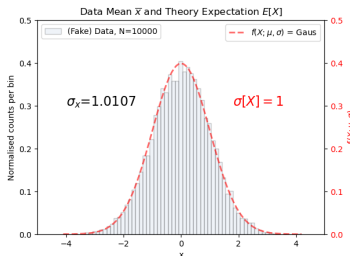
Equivalent in mixed notation $V[X] = E[X^2] - \mu^2$

I think this second form is the most useful formula from this whole week.

Variance is a measure of spread

Variance: amount of spread about the mean value.

Easiest to see this using this form: $V[x] = \frac{1}{N} \sum_i (x_i - \bar{x})^2$.



It is the **Standard Deviation** $\sigma_x = \sqrt{V[x]}$ that we use as our **Statistical Uncertainty** when quoting the mean of the data as $\bar{x} \pm \sigma$.

Aside: Unbiased Variance

The **Unbiased Variance** of a dataset of 1D measurements x is:

$$V[x] = \frac{1}{N-1} \sum_i^N (x_i - \bar{x})^2$$

This has a larger normalisation factor, and we shall see later that this takes care of the problem with calculating the variance from finite-sized dataset. I have provided a (very long) proof of this in an example attached to a later topic Introduction to Estimators.

Put a pin in this for later!

Combining Datasets

So far we have only considered the mean and variance for a single dataset. In real life people collect many datasets of differing size and quality that are in essence measuring the same RV.

To combine these datasets we want to factor in their size and quality, rather than just throw all the measurements into a single uber dataset. To do this we use **Weights**.

The **Weighted Mean** is $\bar{x}_w = \frac{\sum_j^Z x_{(j)} w_j}{\sum_j^Z w_j}$, where $w_j = \frac{1}{V[x_{(j)}]}$

The Weighted Mean

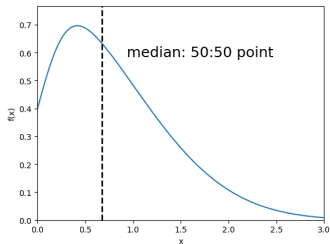
The **Weighted Mean** is $\bar{x}_w = \frac{\sum_j^Z x_{(j)} w_j}{\sum_j^Z w_j}$, where $w_j = \frac{1}{V[x_{(j)}]}$

Note that the sum is over the Z datasets, so the calculation is iterative. First find the mean and variance of each dataset (the variance is needed for the weight) then combine them.

The variance on the weighted mean is no fun at this stage. We will soon see that it can be expressed very simply.

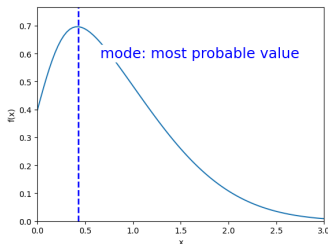
Definitions not used in this module

The **Median** of a dataset (or RV) is the value at which half the measurements lie at or above it.



Definitions not used in this module

The **Mode** of a dataset (or RV) is the most probable value.



Definitions not used in this module

The **RMS** (Root-Mean-Square) of a dataset is

$$\bar{x}_{RMS} = \left(\frac{1}{N} \sum_i^N x_i^2 \right)^{\frac{1}{2}}$$

The RMS is generally used in electronics and sound engineering.

Definitions not used in this module

The **Geometric Mean** of a dataset is

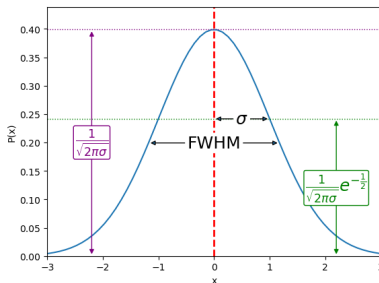
$$\bar{x}_{Geo} = \left(\prod_i^N x_i \right)^{\frac{1}{N}}$$

The Geometric mean is sometimes used in finance.

Definitions not used in this module

The **FWHM** (Full Width at Half Max) of a dataset is

$$\text{FWHM} = 2\sqrt{2\ln 2}\sigma \approx 2.35\sigma$$



Definitions not used in this module

The **Skewness** of a dataset is $\gamma = \frac{1}{N} \sum_0^N \frac{(x - \bar{x})^3}{\sigma^3}$

If the distribution is shifted to positive values, then the skew is positive, and vice versa.

The skewness looks very similar to the variance. It is in fact the **direction of variance**. More about moments [here](#).

Thanks For Your Time

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